THE REDUCED CLOSURE OF THE IMAGE OF A MORPHISM OF GROUPOIDS OVER A PERFECT FIELD IS A GROUPOID SCHEME

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Allowed standard prerequisites (editorial).

- The morphism of groupoids over a field situation fixing R_1, R_2, a and the structure maps (more-groupoids chapter, Situation situation-morphism-groupoids-on-field, canonical tag 04Q5; kind reference). cited at statement lines 1781-1782. Supplies the notation and assumptions of the statement, here with k perfect.
- Definition of the reduced induced scheme structure on a closed subset (Schemes chapter, Definition schemes-definition-reduced-induced-scheme, canonical tag 01J4). cited at statement lines 1784-1785. Fixes the scheme Z as the reduced closed subscheme of R_2 supported on the closure of the image of a.
- A scheme reduced and finite type over a perfect field is geometrically reduced (Varieties chapter, Lemma varieties-lemma-perfect-reduced, canonical tag 020I). cited at proof lines 1795-1798. Gives that Z, reduced over the perfect field k, is geometrically reduced over k via either projection.
- Geometrically reduced schemes stay reduced after any base change (Varieties chapter, Lemma varieties-lemma-geometrically-reduced-any-base-change, canonical tag 035Z). cited at proof lines 1799-1801. Yields that the self fibre product Z x_{s_2, U, t_2} Z, a base change of the geometrically reduced Z, is reduced, hence a closed subscheme of Z.
- Over a field the composition of a groupoid induces a morphism on the reduced closure of the image (more-groupoids chapter, Lemma lemma-map-groupoids-on-field-image, canonical tag 04Q7; also in this lane's queue). cited at proof lines 1802-1805. Applied once Z x_{s_2, U, t_2} Z is reduced, it gives the morphism Z x_{s_2, U, t_2} Z to Z induced by c, the composition arrow of the restricted groupoid.

Source reference. Original reference 04Q5: sources/stacks-chow/tags.txt, source lines 6138–6138. Cited at statement lines 1781-1782: notation and assumptions fixing R_1, R_2, a and the structure maps, with k perfect.

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Source reference. Original reference 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Cited at statement lines 1784-1785: Z is the reduced closed subscheme whose underlying space is the closure of the image of a.

Source reference. Original lemma 020I: sources/stacks-chow/tags.txt, source lines 2635–2635. Cited at proof lines 1795-1798: over the perfect field k the reduced scheme Z is geometrically reduced over k via either projection.

Source reference. Original lemma 035Z: sources/stacks-chow/tags.txt, source lines 4146–4146. Cited at proof lines 1799-1801: the fibre product Z x_{s_2, U, t_2} Z, a base change of the geometrically reduced Z, is reduced.

Source reference. Original lemma 04Q7: sources/stacks-chow/tags.txt, source lines 6140–6140. Cited at proof lines 1802-1805: since Z x_{s_2, U, t_2} Z is reduced, c induces a morphism Z x_{s_2, U, t_2} Z to Z.

Lemma 0.2. *Notation and assumptions as in Situation 04Q5. Assume that k is perfect. Let $Z \subset R_2$ be the reduced closed subscheme (see Schemes, Definition 01J4) whose underlying topological space is the closure of the image of $a : R_1 \to R_2$. Then*

$$(U, Z, s_2|_Z, t_2|_Z, c_2|_Z)$$

is a groupoid scheme over S.

Proof. We first explain why the statement makes sense. Since U is the spectrum of a perfect field k, the scheme Z is geometrically reduced over k (via either projection), see Varieties, Lemma 020I. Hence the scheme $Z \times_{s_2,U,t_2} Z \subset Z$ is reduced, see Varieties, Lemma 035Z. Hence by Lemma 04Q7 we see that c induces a morphism $Z \times_{s_2,U,t_2} Z \to Z$. Finally, it is clear that e_2 factors through Z and that the map $i_2 : R_2 \to R_2$ preserves Z. Since the morphisms of the septuple $(U, R_2, s_2, t_2, c_2, e_2, i_2)$ satisfies the axioms of a groupoid, it follows that after restricting to Z they satisfy the axioms. □